

**CLASS VII MATHS: EXERCISE 5B SOLUTIONS****CHAPTER 5: THE LAWS AND APPLICATIONS OF EXPONENTS**

Complete step-by-step solutions for Multiple Choice, Mental Maths, Case Studies, and Competency Focused Questions.

**Structure of Exercise 5B:**

- **Section 1: Multiple Choice Questions (Page 89)** – Questions 1 to 7
- **Section 2: Mental Maths (Page 89)** – Comparison, Fills, and True/False
- **Section 3: Case Study Based Questions (Page 90)** – Passage I (Depreciation) & Passage II (Population)
- **Section 4: Assertion–Reason Questions (Pages 90–91)** – Questions 1 to 3
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## 1. Multiple Choice Questions (Page 89)

### Question 1: Value of $3^2 + 2^3$

Choose the correct option:  $3^2 + 2^3$  is equal to:

- (a) 15
- (b) 16
- (c) 17
- (d) 18

#### Step-by-step Solution:

$$3^2 = 3 \times 3 = 9$$

$$2^3 = 2 \times 2 \times 2 = 8$$

Summing these values:

$$3^2 + 2^3 = 9 + 8 = 17$$

**Correct Option / Answer:** (c) 17

### Question 2: Value of $(7 - 5)^5$

Choose the correct option:  $(7 - 5)^5$  is equal to:

- (a) 6
- (b) 8
- (c) 16
- (d) 32

**Step-by-step Solution:** First, simplify the base expression inside the parentheses:

$$7 - 5 = 2$$

Now, evaluate the exponential expression:

$$(7 - 5)^5 = 2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$$

**Correct Option / Answer:** (d) 32

**Question 3: Value of  $(-5)^4$** 

Choose the correct option:  $(-5)^4$  is equal to:

- (a) 20
- (b) -125
- (c) 625
- (d) -1024

**Step-by-step Solution:** A negative base raised to an even integer power always results in a positive number:

$$(-5)^4 = (-5) \times (-5) \times (-5) \times (-5) = 25 \times 25 = 625$$

**Correct Option / Answer:** (c) 625

**Question 4: Value of  $(\frac{-3}{5})^5$** 

Choose the correct option:  $(\frac{-3}{5})^5$  is equal to:

- (a)  $\frac{-243}{3125}$
- (b)  $\frac{-81}{625}$
- (c)  $\frac{81}{625}$
- (d)  $\frac{512}{1875}$

**Step-by-step Solution:** Using the power of a quotient rule  $(\frac{a}{b})^n = \frac{a^n}{b^n}$ :

$$\left(\frac{-3}{5}\right)^5 = \frac{(-3)^5}{5^5}$$

Since the exponent is odd, the numerator is negative:

$$\begin{aligned} (-3)^5 &= (-3) \times (-3) \times (-3) \times (-3) \times (-3) = -243 \\ 5^5 &= 5 \times 5 \times 5 \times 5 \times 5 = 3125 \end{aligned}$$

Thus:

$$\left(\frac{-3}{5}\right)^5 = -\frac{243}{3125}$$

**Correct Option / Answer:** (a)  $\frac{-243}{3125}$

**Question 5: Solve for  $x$** 

The value of  $x$  such that  $\left(\frac{3}{7}\right)^3 \times \left(\frac{3}{7}\right)^{-8} = \left(\frac{3}{7}\right)^{2x+3}$  is:

- (a) -4
- (b) -2
- (c) 0
- (d) 1

**Step-by-step Solution:** Applying the Product Law ( $a^m \times a^n = a^{m+n}$ ) on the left-hand side:

$$\left(\frac{3}{7}\right)^3 \times \left(\frac{3}{7}\right)^{-8} = \left(\frac{3}{7}\right)^{3+(-8)} = \left(\frac{3}{7}\right)^{-5}$$

Now equate the left-hand side and right-hand side exponents since the bases are identical:

$$\begin{aligned} -5 &= 2x + 3 \\ 2x &= -5 - 3 \\ 2x &= -8 \\ x &= -4 \end{aligned}$$

**Correct Option / Answer:** (a) -4

**Question 6: Complex nested exponent evaluation**

The value of  $\left[ \left\{ \left( \frac{-1}{3} \right)^{-2} \right\}^{-1} \right]^2$  is:

- (a) 81
- (b) -81
- (c)  $\frac{1}{81}$
- (d)  $-\frac{1}{81}$

**Step-by-step Solution:** Using the Power of a Power Law  $((a^m)^n)^p = a^{m \times n \times p}$ , we can multiply all exponents:

$$\text{Combined exponent} = (-2) \times (-1) \times 2 = 4$$

Therefore:

$$\left[ \left\{ \left( \frac{-1}{3} \right)^{-2} \right\}^{-1} \right]^2 = \left( \frac{-1}{3} \right)^4 = \frac{(-1)^4}{3^4} = \frac{1}{81}$$

**Correct Option / Answer:** (c)  $\frac{1}{81}$

**Question 7: Express 34613000 in exponential form**

The number 34613000 when expressed in exponential form is equal to:

- (a)  $3461.3 \times 10^3$
- (b)  $3.4613 \times 10^7$
- (c)  $0.34613 \times 10^9$
- (d)  $34.613 \times 10^5$

**Step-by-step Solution:** Let's write 34613000 in standard scientific notation. Move the decimal point 7 places to the left:

$$34,613,000 = 3.4613 \times 10^7$$

Let's verify each option:

- (a)  $3461.3 \times 10^3 = 3461.3 \times 1000 = 3,461,300$  (Incorrect)
- (b)  $3.4613 \times 10^7 = 3.4613 \times 10,000,000 = 34,613,000$  (Correct)
- (c)  $0.34613 \times 10^9 = 346,130,000$  (Incorrect)
- (d)  $34.613 \times 10^5 = 3,461,300$  (Incorrect)

**Correct Option / Answer:** (b)  $3.4613 \times 10^7$

## 2. Mental Maths (Page 89)

### Question 1: Which is larger?

Compare the following pairs to determine which number is larger:

- (i)  $2^3$  or  $3^2$
- (ii)  $2^5$  or  $5^2$
- (iii)  $3^7$  or  $7^3$

### Step-by-step Comparison:

- (i) **Compare  $2^3$  and  $3^2$ :**

$$2^3 = 8, \quad 3^2 = 9 \implies 9 > 8$$

So,  $3^2$  is larger.

- (ii) **Compare  $2^5$  and  $5^2$ :**

$$2^5 = 32, \quad 5^2 = 25 \implies 32 > 25$$

So,  $2^5$  is larger.

- (iii) **Compare  $3^7$  and  $7^3$ :**

$$3^7 = 2187, \quad 7^3 = 343 \implies 2187 > 343$$

So,  $3^7$  is larger.

### Question 2: Fill in the blanks

Complete the following mathematical statements:

- (i) In an exponential form  $x^m$ ;  $x$  is called the \_\_\_\_\_.
- (ii) We have:  $\frac{a^m}{a^n} = \frac{1}{a^{n-m}}$  if \_\_\_\_\_.
- (iii) Any number to the power 0 is equal to \_\_\_\_\_.
- (iv) The exponential form is also called \_\_\_\_\_.
- (v) The reciprocal of  $x^a$  is equal to  $x$  to the power \_\_\_\_\_.

### Completed Solutions:

- (i) In  $x^m$ ,  $x$  is called the **base**.
- (ii)  $\frac{a^m}{a^n} = \frac{1}{a^{n-m}}$  if  $m < n$  (where  $a \neq 0$ ).
- (iii) Any non-zero number to the power 0 is equal to **1**.
- (iv) The exponential form is also called **index form** (or **power form / exponential notation**).
- (v) The reciprocal of  $x^a$  is equal to  $x$  to the power  $-a$  (since  $\frac{1}{x^a} = x^{-a}$ ).

**Question 3: True (T) or False (F)**

Determine if the following mathematical statements are true or false:

- (i) For any non-zero number  $a$ , we have  $(a^m)^n = a^{m+n}$
- (ii) If  $a$  and  $b$  are non-zero numbers, then  $\left\{\left(\frac{a}{b}\right)^m\right\}^n = \left(\frac{b}{a}\right)^{-mn}$
- (iii) For a non-zero number  $x$ ;  $(x^m \times x^n)$  is equal to  $x$  to the power  $(m+n)$
- (iv) If a number  $p$  is multiplied  $n$  times, then the resulting number is  $p^n$
- (v) In an exponential notation  $x^n$ ;  $n$  is called the index

**Detailed Solutions:**

- (i) **False.** According to the Power of a Power law,  $(a^m)^n = a^{mn}$ , not  $a^{m+n}$ .
- (ii) **True.** Since  $\left\{\left(\frac{a}{b}\right)^m\right\}^n = \left(\frac{a}{b}\right)^{mn}$ , and taking reciprocal of the base changes the sign of exponent:  $\left(\frac{a}{b}\right)^{mn} = \left(\frac{b}{a}\right)^{-mn}$ .
- (iii) **True.** This is the fundamental Product Law of exponents:  $x^m \times x^n = x^{m+n}$ .
- (iv) **True.** Repeated multiplication is the definition of exponential form:  $\underbrace{p \times p \times \cdots \times p}_{n \text{ times}} = p^n$ .
- (v) **True.** In  $x^n$ ,  $n$  is known interchangeably as the index, power, or exponent.

**Correct Option / Answer:** (i) False (ii) True (iii) True (iv) True (v) True

**3. Case Study Based Questions (Page 90)****Case Study I: Computer Value Depreciation**

**Context:** A computer purchased for **Rs. 72900** loses **two-thirds** of its value every year. Its value is evaluated at the end of every year.

**Note on Depreciation:** If the computer loses  $\frac{2}{3}$  of its value, it retains exactly  $1 - \frac{2}{3} = \frac{1}{3}$  of its value at the end of each year.



**Case Study I – Question 1: Mathematical expression for value after  $n$  years**

Which of the following expressions gives the value of the computer (in Rs.) after  $n$  years?

- (a)  $\frac{72900}{\left(\frac{2}{3}\right)^n}$
- (b)  $\frac{2^n \times 27900}{3^n}$
- (c)  $\frac{72900}{3^n}$
- (d)  $\frac{72900}{2^n \times 3^n}$

Let the initial value of the computer be  $V_0 = 72900$ . At the end of year 1, the value is  $V_1 = V_0 \times \frac{1}{3}$ . At the end of year  $n$ , the value  $V_n$  is:

$$V_n = 72900 \times \left(\frac{1}{3}\right)^n = \frac{72900}{3^n}$$

**Correct Option / Answer:** (c)  $\frac{72900}{3^n}$

**Case Study I – Question 2: Value after 3 years**

Find the value of the computer after 3 years:

- (a) Rs. 8100
- (b) Rs. 2430
- (c) Rs. 5600
- (d) Rs. 2700

Substitute  $n = 3$  into our formula:

$$V_3 = \frac{72900}{3^3} = \frac{72900}{27} = 2700$$

The value of the computer after 3 years will be Rs. 2700.

**Correct Option / Answer: (d) Rs. 2700**

**Case Study I – Question 3: Value drops below Rs. 200**

In how many years will the value of the computer be less than Rs. 200?

- (a) 6 years
- (b) 7 years
- (c) 8 years
- (d) 10 years

Let's compute the value of the computer year-by-year:

- Year 0 (Initial): Rs. 72900
- Year 1:  $\frac{72900}{3} = \text{Rs. } 24300$
- Year 2:  $\frac{24300}{3} = \text{Rs. } 8100$
- Year 3:  $\frac{8100}{3} = \text{Rs. } 2700$
- Year 4:  $\frac{2700}{3} = \text{Rs. } 900$
- Year 5:  $\frac{900}{3} = \text{Rs. } 300$
- Year 6:  $\frac{300}{3} = \text{Rs. } 100$

Since Rs. 100 is less than Rs. 200, the value falls below Rs. 200 after **6 years**.

**Correct Option / Answer: (a) 6 years**

**Case Study I – Question 4: Total depreciation in 4 years**

By how much will the value of the computer reduce in 4 years?

- (a) Rs. 24300
- (b) Rs. 1800
- (c) Rs. 8100
- (d) Rs. 72000

The reduction in value is the initial price minus the value at year 4:

$$\text{Value after 4 years } V_4 = \frac{72900}{3^4} = \frac{72900}{81} = \text{Rs. } 900$$

$$\text{Total Depreciation} = V_0 - V_4 = 72900 - 900 = \text{Rs. } 72000$$

**Correct Option / Answer:** (d) Rs. 72000

**Case Study II: Bacterial Population Growth**

**Context:** In a bacteria culture under observation in a laboratory, the population of **50 bacteria** doubles itself every hour.

**Growth rate:** The base is 2 (since it doubles), and the initial population is  $P_0 = 50$ .

**Case Study II – Question 1: Mathematical expression after  $n$  hours**

Which of the following expressions gives the bacterial population after  $n$  hours?

- (a)  $\frac{50}{2^n}$
- (b)  $25^n$
- (c)  $50 \times 2^n$
- (d)  $\frac{50^n}{2}$

Since the population starts at 50 and multiplies by 2 every hour:

$$P_n = 50 \times 2^n$$

**Correct Option / Answer:** (c)  $50 \times 2^n$

**Case Study II – Question 2: Population after 3 hours**

The population size of the bacteria after 3 hours will be:

- (a) 200
- (b) 300
- (c) 400
- (d) 500

Substitute  $n = 3$  into our formula:

$$P_3 = 50 \times 2^3 = 50 \times 8 = 400$$

**Correct Option / Answer: (c) 400**

**Case Study II – Question 3: Population after 1 day**

How many bacteria will be there in the culture after 1 day?

- (a)  $\frac{50}{2^{12}}$
- (b)  $50 \times 2^{24}$
- (c)  $50 \times 2^{12}$
- (d)  $\frac{50}{2^{24}}$

There are exactly 24 hours in 1 day. Let  $n = 24$ :

$$P_{24} = 50 \times 2^{24}$$

**Correct Option / Answer: (b)  $50 \times 2^{24}$**

**Case Study II – Question 4: Hours required to exceed 1000 bacteria**

Find the number of hours after which the population size of the bacteria will be larger than 1000:

- (a) 5
- (b) 6
- (c) 8
- (d) 10

Let's compute the population hour-by-hour:

- Hour 0: 50
- Hour 1:  $50 \times 2 = 100$
- Hour 2:  $100 \times 2 = 200$
- Hour 3:  $200 \times 2 = 400$
- Hour 4:  $400 \times 2 = 800$
- Hour 5:  $800 \times 2 = 1600$

Since 1600 is greater than 1000, the population exceeds 1000 after exactly **5 hours**.

**Correct Option / Answer: (a) 5**

## 4. Assertion–Reason Questions (Pages 90–91)

## Assertion-Reason Instructions

In the following questions, a statement of Assertion (A) is followed by a statement of Reason (R). Choose:

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (c) (A) is true but (R) is false.
- (d) (A) is false but (R) is true.

## Question 1

**Assertion (A):**  $\frac{x^4}{y^3}$  can also be written as  $\frac{x+x+x+x}{y+y+y}$ .

**Reason (R):** In  $x^m$ ,  $x$  is called the base and  $m$  is called the exponent.

**Analysis:**

- **Assertion (A) is false:**  $x^4$  is repeated multiplication ( $x \times x \times x \times x$ ), not repeated addition ( $x + x + x + x$ ). Similarly,  $y^3 = y \times y \times y$ . Therefore,  $\frac{x^4}{y^3} \neq \frac{4x}{3y}$ .
- **Reason (R) is true:** By definition, in any exponential term  $x^m$ ,  $x$  is the base and  $m$  is the index/exponent.

**Correct Option / Answer:** (d) (A) is false but (R) is true.

## Question 2

**Assertion (A):**  $x^m \div y^m = (x \div y)^m$ **Reason (R):**  $a^m \times b^m = (ab)^m$ **Analysis:**

- **Assertion (A) is true:** The quotient of powers with same exponents is indeed  $x^m \div y^m = \left(\frac{x}{y}\right)^m$ .
- **Reason (R) is true:** The product of powers with same exponents is  $a^m \times b^m = (ab)^m$ .
- **Relationship:** Although both laws are correct and represent twin properties of powers with identical exponents, the multiplication law (Reason) is not a direct algebraic derivation/explanation of the division law (Assertion). Hence, (R) is not the correct explanation of (A).

**Correct Option / Answer:** (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).

## Question 3

**Assertion (A):**  $2^0 + 3^0 + 4^0 = (2 + 3 + 4)^0$

**Reason (R):**  $x^0 = 1$ .

**Analysis:**

- **Assertion (A) is false:** Let's evaluate both sides using the zero-power law:

$$\text{LHS} = 2^0 + 3^0 + 4^0 = 1 + 1 + 1 = 3$$

$$\text{RHS} = (2 + 3 + 4)^0 = 9^0 = 1$$

Since  $3 \neq 1$ , Assertion (A) is incorrect.

- **Reason (R) is true:** Any non-zero base raised to power 0 is exactly 1.

**Correct Option / Answer:** (d) (A) is false but (R) is true.

## 5. Competency Focused Questions (Pages 91–92)

Question 1: Value of  $2^3 + 2^3 + 2^3 + 2^3$ 

Choose the correct option:  $2^3 + 2^3 + 2^3 + 2^3$  is equal to:

- (a)  $2^5$
- (b)  $2^{12}$
- (c) 281
- (d) 216

This represents repeated addition of the term  $2^3$ , which is done 4 times:

$$2^3 + 2^3 + 2^3 + 2^3 = 4 \times 2^3$$

Write 4 as an exponential term with base 2 (i.e.,  $4 = 2^2$ ):

$$4 \times 2^3 = 2^2 \times 2^3$$

Apply the Product Law of exponents ( $a^m \times a^n = a^{m+n}$ ):

$$2^2 \times 2^3 = 2^{2+3} = 2^5$$

**Correct Option / Answer:** (a)  $2^5$



**Question 2: Rational number expression**

The value of  $\frac{\left(\frac{-1}{2}\right)^5}{\left(\frac{-1}{2}\right)^4} \div \frac{\left(\frac{-1}{8}\right)}{\left(\frac{-1}{4}\right)}$  is:

- (a) 2
- (b) 0
- (c) 1
- (d) -1

Let's simplify both fractions independently:

1. **First term:** Apply the Quotient Law of exponents  $\frac{a^m}{a^n} = a^{m-n}$ :

$$\frac{\left(\frac{-1}{2}\right)^5}{\left(\frac{-1}{2}\right)^4} = \left(\frac{-1}{2}\right)^{5-4} = \left(\frac{-1}{2}\right)^1 = -\frac{1}{2}$$

2. **Second term:**

$$\frac{\left(\frac{-1}{8}\right)}{\left(\frac{-1}{4}\right)} = \left(-\frac{1}{8}\right) \times \left(-\frac{4}{1}\right) = \frac{4}{8} = \frac{1}{2}$$

Now execute the division between the two terms:

$$\left(-\frac{1}{2}\right) \div \left(\frac{1}{2}\right) = \left(-\frac{1}{2}\right) \times \left(\frac{2}{1}\right) = -1$$

**Correct Option / Answer: (d) -1**

**Question 3: Algebraic exponent cancellation**

The value of  $\left(\frac{a^p}{a^q}\right) \times \left(\frac{a^q}{a^r}\right) \times \left(\frac{a^r}{a^p}\right)$  is:

- (a) 1
- (b)  $a^{pqr}$
- (c)  $\frac{1}{a^{pqr}}$
- (d)  $a$

Apply the laws of exponents step-by-step:

$$\text{Expression} = a^{p-q} \times a^{q-r} \times a^{r-p}$$

Add the exponents using the Product Law:

$$= a^{(p-q)+(q-r)+(r-p)} = a^{p-q+q-r+r-p} = a^0$$

Since any non-zero number to power 0 is 1:

$$a^0 = 1$$

**Correct Option / Answer: (a) 1**

**Question 4: Value of  $a^b - b^a$** 

The value of  $a^b - b^a$ , if  $a = 3$  and  $b = 7$  is:

- (a) 1825
- (b) 1840
- (c) 1844
- (d) 1850

Substitute the given values into the expression:

$$a^b - b^a = 3^7 - 7^3$$

Calculate the values:

$$3^7 = 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 = 2187$$

$$7^3 = 7 \times 7 \times 7 = 343$$

Now subtract the results:

$$2187 - 343 = 1844$$

**Correct Option / Answer: (c) 1844**

**Question 5: Simplification of complex fraction**

The value of  $\frac{\frac{3^4}{3^3} + \left(\frac{1}{9} \times 4\right)^2}{\left(\frac{-1}{3}\right)^3 \div \left(\frac{1}{-2}\right)^3}$  is:

- (a)  $\frac{81}{24}$
- (b)  $\frac{259}{24}$
- (c)  $\frac{81}{8}$
- (d)  $\frac{259}{8}$

**Step-by-step Calculation:**

1. **Numerator first term:**

$$\frac{3^4}{3^3} = 3^{4-3} = 3^1 = 3$$

*Note: If the book typo reads  $3^4/2^3 = 81/8$ , the final output becomes 6689/192 (not in options). Thus, the correct intended math expression is indeed  $3^4/3^3 = 3$ .*

2. **Numerator second term:**

$$\left(\frac{1}{9} \times 4\right)^2 = \left(\frac{4}{9}\right)^2 = \frac{16}{81}$$

3. **Combine Numerator:**

$$\text{Numerator} = 3 + \frac{16}{81} = \frac{3 \times 81 + 16}{81} = \frac{243 + 16}{81} = \frac{259}{81}$$

4. **Denominator:**

$$\left(\frac{-1}{3}\right)^3 \div \left(\frac{1}{-2}\right)^3 = \frac{-1}{27} \div \frac{-1}{8} = \frac{-1}{27} \times \frac{-8}{1} = \frac{8}{27}$$

5. **Final Division:**

$$\frac{\text{Numerator}}{\text{Denominator}} = \frac{259}{81} \div \frac{8}{27} = \frac{259}{81} \times \frac{27}{8} = \frac{259}{3 \times 8} = \frac{259}{24}$$

**Correct Option / Answer:** (b)  $\frac{259}{24}$

**Question 6: Repeated negative multiplication**

The value of  $(-8) \times (-8) \times (-8) \times (-8) \times (-8) \times (-8) \times (-8) \times (-8)$  is:

- (a)  $-2(8)^5$
- (b)  $-(2)^{16}$
- (c)  $-(4)^8$
- (d) none of these

Let's count the number of factors of  $(-8)$ : We have exactly 8 factors of  $(-8)$ . Thus, the expression in exponential form is:

$$(-8)^8$$

Since the exponent (8) is an even integer, the result is positive:

$$(-8)^8 = 8^8 = (2^3)^8 = 2^{24} = 16,777,216$$

Let's evaluate the given options:

- (a)  $-2(8)^5 = -2(2^3)^5 = -2 \times 2^{15} = -2^{16}$  (Negative)
- (b)  $-(2)^{16} = -2^{16}$  (Negative)
- (c)  $-(4)^8 = -(2^2)^8 = -2^{16}$  (Negative)

Since  $2^{24}$  is positive and much larger than  $-2^{16}$ , none of the choices (a), (b), or (c) are correct.

**Correct Option / Answer:** (d) none of these

**Question 7: Product & Quotient Law combination**

The value of  $(p^x \times p^y) \div (p^x \div p^y)$  is:

- (a)  $2p^x$
- (b)  $p^{2x}$
- (c)  $2p^y$
- (d)  $p^{2y}$

Apply the exponent laws to the terms inside the parentheses:

$$\text{Numerator (Product Law): } p^x \times p^y = p^{x+y}$$

$$\text{Denominator (Quotient Law): } p^x \div p^y = p^{x-y}$$

Now perform the division using the Quotient Law:

$$\frac{p^{x+y}}{p^{x-y}} = p^{(x+y)-(x-y)} = p^{x+y-x+y} = p^{2y}$$

**Correct Option / Answer:** (d)  $p^{2y}$

**Question 8: Multi-variable evaluation**

The value of  $x^y - y^x + xy$ , if  $x = 6, y = 2$  is:

- (a) -12
- (b) 12
- (c) -16
- (d) 16

Substitute  $x = 6$  and  $y = 2$  directly:

$$x^y - y^x + xy = 6^2 - 2^6 + (6 \times 2)$$

Calculate each term:

$$6^2 = 36$$

$$2^6 = 64$$

$$6 \times 2 = 12$$

Combine the values:

$$36 - 64 + 12 = -28 + 12 = -16$$

**Correct Option / Answer: (c) -16**

**Question 9: Rational exponent ratio**

If  $3^x = 500$ , then the value of  $3^{x-2}$  is:

- (a)  $\frac{500}{9}$
- (b)  $\frac{1000}{9}$
- (c)  $\frac{500}{3}$
- (d)  $\frac{100}{9}$

Rewrite the expression  $3^{x-2}$  using exponent subtraction law in reverse (Quotient Law):

$$3^{x-2} = \frac{3^x}{3^2}$$

Substitute  $3^x = 500$ :

$$3^{x-2} = \frac{500}{3^2} = \frac{500}{9}$$

**Correct Option / Answer:** (a)  $\frac{500}{9}$



**Question 10: Conjecture proof / disproof**

Rajat claims, "A negative number raised to a power is always less than the number itself." Rajat's statement is:

- (a) true
- (b) false
- (c) can't say anything
- (d) none of these

Let's test Rajat's statement with a negative number, say  $a = -2$ :

- For  $n = 1$ :  $(-2)^1 = -2$  (Equal to the number itself)
- For  $n = 2$  (Even):  $(-2)^2 = 4$  (**Greater** than the number itself, since  $4 > -2$ )
- For  $n = 3$  (Odd):  $(-2)^3 = -8$  (Less than the number itself)

Since  $(-2)^2 = 4$  is greater than  $-2$ , a negative number raised to an even power is positive, and thus is always greater than the negative base. Hence, Rajat's statement is false.

**Correct Option / Answer: (b) false**

**Question 11: Simplify  $\{(9^2)^4 \times 9^5\} \div 9^8$** 

$\{(9^2)^4 \times 9^5\} \div 9^8$  is equal to:

- (a) 9
- (b)  $9^8$
- (c)  $9^5$
- (d)  $9^9$

Apply exponent laws:

1. Power of a Power:  $(9^2)^4 = 9^{2 \times 4} = 9^8$ .
2. Product Law inside curly brackets:  $9^8 \times 9^5 = 9^{8+5} = 9^{13}$ .
3. Quotient Law for final division:  $9^{13} \div 9^8 = 9^{13-8} = 9^5$ .

**Correct Option / Answer: (c)  $9^5$**